DO NOW — RECALL

Lesson 105 · RAM, ROM and Virtual Storage: Different R

Name: _____ Class: _____ Date: _____

1. Where are running program instructions and data commonly held?

2. What does ROM or firmware flash retain?

3. What is a remote network drive an example of?

4. Rising edge D-type flip-flop D0 Q1: Q just after edge?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

RAM, plus CPU caches/registers.

2. Model answer

Startup/firmware instructions without power.

3. Model answer

Virtual storage.

4. Model answer

0, because D is sampled at the active edge.

By the end, you can…

• RAM holds currently used programs/data and normally loses content without power; ROM or firmware flash retains startup routines.

• OCR virtual storage means logical storage presented from physically remote or pooled devices, such as a network drive or cloud service.

• Do not confuse virtual storage with virtual memory. Virtual memory uses secondary storage to extend the apparent addressable memory of processes; virtual storage is remote-file presentation.

Vocabulary — know these terms

RAM — Volatile working memory.

ROM — Non-volatile storage for relatively fixed code.

virtual storage — Remote/pooled physical storage presented logically.

virtual memory — OS memory abstraction using paging.

NEW LEARNING

Read, annotate and explain

Understand

RAM holds currently used programs/data and normally loses content without power; ROM or firmware flash retains startup routines.

Apply

OCR virtual storage means logical storage presented from physically remote or pooled devices, such as a network drive or cloud service.

Evaluate

Do not confuse virtual storage with virtual memory. Virtual memory uses secondary storage to extend the apparent addressable memory of processes; virtual storage is remote-file presentation.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A pupil opens a document from a mounted school network drive. Where is the file physically stored?

Explain the choice

On one or more remote network storage devices, even if Finder/Explorer shows a drive like local storage.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain the role and volatility of RAM and ROM/firmware storage. [6]

Question 2

Define virtual storage and give an example, two benefits and a disadvantage. [8]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Correct a pupil's claim that 'virtual storage is extra RAM created when memory fills up'. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

OCR H446 1.1.3(d) virtual storage is another term for swapping RAM pages.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.